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LINUX ASSIGNMENT – 1

1) What If There Were No Operating System?

OS – Operating system acts as a bridge at both software and hardware this is the way to which interacts to the users and devices.

It controls:

- Process management
- Memory management
- Device management
- Security management
- Input/Output device controls

What if there were no “OS”

We can't run applications

Without operating system not able to interact with hardware devices

Can't able to organize and save file and folders

The system can only run one built-in set of instructions at a time

We can't able to perform multitasking.

There is no network communication

Can't able to use graphic cards

S no	Features	Without OS	With OS
1	Networking	We want to manually build our TCP/IP layers	It already built in the network sockets
2	Memory Space	Without OS the application have direct access to the physical RAM	It was handled by separate virtual memory sandbox
3	System errors	Collision will be in the hardware devices that cannot perform multi-tasking	This have Unified kernel logs to track the faults.

2)

Sno	Different Roles	Linux Distribution	Cyber security related function
1	Penetration Testing	Kali / parrot OS	It comes ready out-of-the-box with the special software tools needed to crack passwords using your computer's graphics card
2	Digital Forensics	CAINE	It automatically locks hard drives into a read – only mode.
3	Lab Threat Monitoring	Security Onion	It takes many different complicated tools that watch internet traffic and combines all of their data into one easy-to-read screen
4	Anonymity & OSINT	Tails	preventing outside entities from tracking the lab's public IP.
5	Target Infrastructure	Alpine Linux	easily controlled footprint to test microservice security and compile lightweight malware samples.

3)

```
siva@linux:~$ ls /home
siva
siva@linux:~$ echo $
$
siva@linux:~$ ping -c 1 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.128 ms

--- 127.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.128/0.128/0.128/0.000 ms
siva@linux:~$ cat non_existent_file.txt
cat: non_existent_file.txt: No such file or directory
siva@linux:~$ apt-get update
Reading package lists... Done
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
siva@linux:~$
```

Commands:

- 1) ls /home – success
- 2) ping -c 1 127.0.0.1 – success (used for network is active or not)
- 3) apt-get update – fail (Permission denied)
- 4) grep --invalid-option – fail (incorrect command syntax)
- 5) cat non_existent_file.txt – fail (file or directory not found)